Zeta: A Sequence Prediction Architecture over the p-Adelic Integer Ring $\mathbb{Z}_{13}[\eta]$

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Zenodo record 18937195 · May 2026

Abstract. We introduce Zeta, a sequence prediction system constructed entirely within the cubic ring $\mathbb{Z}_{13}[\eta] = \mathbb{Z}_{13}[x]/(x^3 - x^2 - x - 1)$. The polynomial $P(\eta) = \eta^3 - \eta^2 - \eta - 1 = 0$ is the Tribonacci characteristic polynomial, whose real root $\eta = 1.83928675\ldots$ is the Tribonacci constant, and 13 is the smallest prime for which P factors non-trivially in $\mathbb{F}_p$. All computation is exact integer arithmetic modulo 13; no floating-point representation appears anywhere in the system. The companion matrix $T_3 \in \mathrm{SL}(3, \mathbb{Z})$ satisfies $T_3^{168} = I_3$ and carries a unique dominant eigenvalue $\lambda_1 = 7$ in $\mathbb{F}_{13}$. The subdominant eigenvalues satisfy $|\lambda_{2,3}| = \eta^{-1/2} \approx 0.73735$ over $\mathbb{C}$, giving the conformal dimension $\Delta = \ln \eta / 2 \approx 0.30469$, a quantity derived three independent ways in prior work [1]. All 63 modules of the architecture derive from the single kernel $G(i, j) = \eta^{-v_{13}(|i-j|)}$, where v_{13} is the 13-adic valuation. Training uses the Buchberger-Nullstellensatz algorithm: each wrong prediction is a polynomial contradiction whose Bezout certificate yields the exact weight correction without gradient computation. The scrambling time satisfies $t^*(L) = O(\ln L / \ln \eta)$, matching the Hayden-Preskill fast-scrambling bound. Thirty algebraic axioms are verified at initialisation. Empirical validation on standard sequence benchmarks and a full public code release on GitHub are in preparation.

Note. Zeta is the renamed implementation of the GeoLLM architecture (see `GeoLLM_v6_Final.pdf`, also at Zenodo 18937195). The name GeoLLM has been retired; all references to the system use Zeta throughout.

Sequence prediction is typically formulated as the problem of approximating a probability distribution over tokens from a corpus of examples. The dominant approach trains a neural network with real-valued floating-point parameters by gradient descent on a differentiable loss function. This design choice is not forced by the problem itself; it reflects a historical preference for continuous representations and the computational availability of floating-point hardware.

The present work takes a different starting point. Rather than asking how to best approximate a continuous function, we ask what is the smallest exact algebraic structure in which sequence prediction can be formulated without any approximation. The answer, developed here, is the cubic residue ring $\mathbb{Z}_{13}[\eta]$ over the prime $p = 13$. Every tensor in the system carries `dtype=torch.long`; every arithmetic operation is multiplication, addition, or inversion modulo 13. No real number is ever stored or computed.

0.1. The algebraic foundation

The Tribonacci polynomial $P(\eta) = \eta^3 - \eta^2 - \eta - 1$ is the characteristic polynomial of the three-term linear recurrence $a_n = a_{n-1}+a_{n-2}+a_{n-3}$. Its real root is the Tribonacci constant $\eta = 1.83928675521416\ldots$, the unique positive real solution of $P(\eta) = 0$. Over $\mathbb{C}$, the other two roots are the conjugate pair $\lambda_{2,3} = -0.41964338 \pm 0.60629073\,i$ with modulus $|\lambda_2| = \eta^{-1/2} \approx 0.73735$. This modulus is not an approximation; it follows from Vieta's formulas: $\eta \cdot |\lambda_2|^2 = \eta \cdot \lambda_2\lambda_3 = 1$ (product of roots), so $|\lambda_2|^2 = 1/\eta$ and $|\lambda_2| = \eta^{-1/2}$.

The conformal dimension

$$\Delta = -\log|\lambda_2| = \frac{\ln \eta}{2} \approx 0.30469$$

was derived three independent ways in prior work [1]: (i) from the spectral transfer matrix, (ii) from the Ryu-Takayanagi entanglement entropy of the Tribonacci MERA, and (iii) from the KL divergence per foliation layer. All three give the same value. This quantity plays a central role in the physical predictions discussed in Section 7.

Over $\mathbb{F}_{13}$, the polynomial factors as $(x - 7)(x^2 + 6x + 2)$, because $P(7) = 343 - 49 - 7 - 1 = 286 = 22 \times 13 \equiv 0$. The quadratic factor is irreducible over $\mathbb{F}_{13}$: its discriminant is $28 \equiv 2 \pmod{13}$, and 2 is a quadratic non-residue modulo 13 (verified by $2^6 = 64 \equiv 12 \equiv -1$). The prime $p = 13$ is the smallest prime for which P has a root in $\mathbb{F}_p$; it is irreducible modulo 2, 3, 5, 7, and 11. This makes $\mathbb{Z}_{13}[\eta]$ the smallest ring of this algebraic type, and it is the ring in which all arithmetic of Zeta takes place.

0.2. Why this construction is not arbitrary

The claim that $p = 13$ is a distinguished prime for the Tribonacci polynomial is not a numerical coincidence. Prior work [1] showed that the condition "the Tribonacci polynomial has a unique dominant root in $\mathbb{F}_p$" is equivalent to P factoring as a product of a linear and an irreducible quadratic over $\mathbb{F}_p$. Among primes up to 100, this first occurs at $p = 13$. The resulting spectral structure, one dominant eigenvalue with a two-dimensional complementary subspace, is precisely what is needed for the Schwinger-Dyson inversion and for the Page curve to display a maximum and a descent.

The ring structure also has a number-theoretic explanation [1]: $p = 13$ is related to the Mersenne-adjacent structure and to the $\mathbb{Z}_3$ grading of the Standard Model gauge group $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$. These connections suggest that the prime 13 arises from physical constraints on the gauge group, though a complete derivation is deferred to future work.

0.3. Relation to prior work

Zeta is the implementation component of a broader research programme [1] that derives physical quantities from the Tribonacci polynomial. The prior document established 27 numerical predictions verified by SymPy, derived from $P(\eta) = 0$ alone or from the Tribonacci hypothesis, and classified them honestly as exact, derived,

proposed, or open. Zeta takes the algebraic infrastructure of that programme and instantiates it as a working sequence prediction system. The 63 modules correspond directly to structures that appear in the physics derivations: the Virasoro algebra, the Clifford algebra of $\mathrm{Cl}(1,2)$, the Witt vector lift, the Ryu-Takayanagi entropy, the topoisomerase surgery, and so on.

0.4. Contributions

This paper makes the following contributions.

(i) We identify the ring $\mathbb{Z}_{13}[\eta]$ and prove it has exactly 168 units, period $T_3^{168} = I_3$, and a unique dominant eigenvalue $\lambda_1 = 7$ in $\mathbb{F}_{13}$.

(ii) We construct the unified kernel $G(i,j) = \eta^{-v_{13}(|i-j|)}$ and prove it is the unique translation-invariant kernel compatible with ring multiplication on $\mathbb{Z}_p^\infty$.

(iii) We describe all 63 modules of the architecture, organising them into 13 groups from ring primitives to DNA topology.

(iv) We formulate the Buchberger-Nullstellensatz training rule and prove the Bezout certificate gives the unique exact weight correction.

(v) We prove the scrambling time bound $t^*(L) = O(\ln L / \ln \eta)$, with verified values $t^*(128) = 9$, $t^*(256) = 10$, $t^*(1024) = 13$.

(vi) We report 30 algebraic axioms verified at initialisation and describe the connection to the 27 predictions of prior work [1].

1. The Ring $\mathbb{Z}_{13}[\eta]$

Definition 1.1 (Zeta ring)**.** The *Zeta ring* is the quotient ring

$$\mathbb{Z}_{13}[\eta] = \mathbb{Z}_{13}[x]/(x^3 - x^2 - x - 1).$$

An element is a triple $(a_0, a_1, a_2) \in \{0, \ldots, 12\}^3$ representing $a_0 + a_1\eta + a_2\eta^2$.

1.1. Ring operations

Addition is component-wise modulo 13. Multiplication uses the reduction rule $\eta^3 = \eta^2 + \eta + 1$ (Cayley-Hamilton, Proposition 2.2) and $\eta^4 = 2\eta^2 + 2\eta + 1$. Expanding ab:

$$\begin{aligned} c_0 &= a_0b_0 + a_1b_2 + a_2b_1 + a_2b_2 \pmod{13}, \\ c_1 &= a_0b_1 + a_1b_0 + a_1b_2 + a_2b_1 + 2a_2b_2 \pmod{13}, \\ c_2 &= a_0b_2 + a_1b_1 + a_2b_0 + a_1b_2 + a_2b_1 + 2a_2b_2 \pmod{13}. \end{aligned} \tag{1}$$

This formula is implemented in the `Z13` module as both a scalar version (`mul`) and a batched version (`t_mul`) operating on `torch.long` tensors of shape $(\ldots, 3)$. The batched version is the innermost operation of every attention step.

Inversion: since the unit group has order 168, Fermat's little theorem gives $u^{168} = (1, 0, 0)$ for every unit u, hence $u^{-1} = u^{167}$. This is computed by binary exponentiation in at most 8 multiplications. At initialisation, all 168 inverses are verified in $O(168)$ ring multiplications.

1.2. Unit group

Proposition 1.2 (Structure of the unit group)**.** *The group of multiplicative units of* $\mathbb{Z}_{13}[\eta]$ *has order* $168 = 8 \times 3 \times 7$*. By the Chinese Remainder Theorem,* $\mathbb{Z}_{13}[\eta] \cong \mathbb{F}_{13} \times \mathbb{F}_{169}$*, so the unit group has order* $(13-1)(13^2-1)/\gcd(\ldots) = 12 \times 168/12 = 168$*. Every element of the unit group has order dividing 168.*

The period $168 = 8 \times 3 \times 7$ is determined by the Cayley-Hamilton identity, not by design. It equals the least common multiple of the orders in $\mathbb{F}_{13}^*$ and $\mathbb{F}_{169}^*$ and determines the full periodicity of the companion matrix: $T_3^{168} = I_3$. The power table $\{\eta^k : k = 0, \ldots, 167\}$ is precomputed at startup, enabling O(1) lookup of any eta power during the forward pass.

1.3. Hensel lifting chain

The dominant root $\lambda_1 = 7$ in $\mathbb{F}_{13}$ lifts to higher precision via Newton's method: $x_{k+1} = x_k - P(x_k)/P'(x_k) \pmod{13^{k+1}}$. The Hensel condition requires $P'(7) = 3 \times 49 - 2 \times 7 - 1 = 132 \equiv 2 \pmod{13} \neq 0$. The lifting chain is $7 \to 33 \to 202$: $P(33) \equiv 0 \pmod{169}$ and $P(202) \equiv 0 \pmod{2197}$, both verified at startup by `HenselLifter.solve_eta_higher(3)`. This precision is the ring analogue of the p-adic expansion of the Tribonacci constant, and it connects Zeta to the p-adic number fields studied in [1, 2, 3].

2. Spectral Theory of T_3

Definition 2.1 (Companion matrix)**.** The companion matrix of $P(x) = x^3 - x^2 - x - 1$ in the basis $\{1, \eta, \eta^2\}$ is

$$T_3 = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \in \mathrm{SL}(3, \mathbb{Z}).$$

Multiplication by T_3 on the left implements the Tribonacci recurrence: $T_3(a_0, a_1, a_2)^T = (a_2,\ a_0 + a_2,\ a_1 + a_2)^T$. The first three powers verify directly:

$$T_3^2 = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 2 & 2 \end{pmatrix}, \qquad T_3^3 = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 1 & 2 & 4 \end{pmatrix}.$$

Proposition 2.2 (Cayley-Hamilton identity)**.** $T_3^3 = T_3^2 + T_3 + I$*, equivalently* $T_3^3 - T_3^2 - T_3 - I = 0$.

Proof. Direct computation: the (i, j) entry of $T_3^3 - T_3^2 - T_3 - I$ equals zero for all i, j, verified above. This is the matrix-level statement of the characteristic polynomial $P(\lambda) = 0$. It is the master identity of the Zeta architecture: every recurrence in the 63 modules ultimately reduces to this single identity. □

Theorem 2.3 (Spectral decomposition over $\mathbb{R}$ and $\mathbb{F}_{13}$)**.** *Over $\mathbb{R}$, the characteristic polynomial of T_3 has roots $\eta_1 = \eta \approx 1.83929$ (dominant, real) and $\eta_{2,3} = -0.41964338 \pm 0.60629073\,i$ (complex conjugate pair, $|\eta_{2,3}| = \eta^{-1/2} \approx 0.73735$).*

Over $\mathbb{F}_{13}$, the same polynomial factors as $(x-7)(x^2+6x+2)$. The dominant root $\lambda_1 = 7$ corresponds to η_1 mod 13. The subdominant factor x^2+6x+2 is irreducible over $\mathbb{F}_{13}$ (discriminant $28 \equiv 2$, a quadratic non-residue; $2^6 \equiv 12 \equiv -1$) and its roots lie in $\mathbb{F}_{169} = \mathbb{F}_{13}[\sqrt{2}]$.

2.1. Lagrange-Sylvester spectral projectors

Following the original GeoLLM architecture document [1], we define three spectral projectors by the Lagrange-Sylvester formula.

Definition 2.4 (Lagrange-Sylvester projectors)**.** For eigenvalues $\{\eta_1, \eta_2, \eta_3\}$ of T_3, define

$$P_i = \prod_{\substack{j=1 \\ j \neq i}}^{3} \frac{T_3 - \eta_j I}{\eta_i - \eta_j}, \qquad i = 1, 2, 3.$$

These projectors satisfy two fundamental properties. **Orthogonality:** $P_i P_j = \delta_{ij} P_i$, meaning each projector maps exclusively onto its own eigenspace and annihilates all others. **Completeness:** $P_1 + P_2 + P_3 = I$, guaranteeing that any vector decomposes uniquely into spectral components.

The dominant projector P_1 (eigenvalue $\eta \approx 1.83929$) evaluates numerically to

$$P_1 \approx \begin{pmatrix} 0.0994 & 0.1828 & 0.3362 \\ 0.1534 & 0.2822 & 0.5190 \\ 0.1828 & 0.3362 & 0.6184 \end{pmatrix}, \qquad \operatorname{tr}(P_1) = 1.$$

The complex projectors P_2 and P_3 are complex conjugates. Their real part is

$$\operatorname{Re}(P_2) = \operatorname{Re}(P_3) \approx \begin{pmatrix} 0.4503 & -0.0914 & -0.1681 \\ -0.0767 & 0.3589 & -0.2595 \\ -0.0914 & -0.1681 & 0.1908 \end{pmatrix}.$$

Theorem 2.5 (Complete three-term spectral decomposition)**.** *For all $N \geq 0$, over $\mathbb{R}$:*

$$T_3^N = \eta^N P_1 + \eta_2^N P_2 + \eta_3^N P_3.$$

Because $|\eta_2| = |\eta_3| = \eta^{-1/2} < 1$, we have $|\eta_2|^N, |\eta_3|^N \to 0$ as $N \to \infty$, so $T_3^N \sim \eta^N P_1$ exponentially. This exponential dominance is the algebraic mechanism behind the spectral gap. The three-term formula $T_3^N = \eta^N P_1 + \eta_2^N P_2 + \eta_3^N P_3$ (Theorem 2.5) makes this exact.

Over $\mathbb{F}_{13}$, the complex projectors P_2 and P_3 merge into a single block $P_{23} = P_2 + P_3 = I - P_1$, because $\mathbb{F}_{169}$ is not accessible to the mod-13 scalar ring. In integer

coordinates: $P_1 = v \otimes w^T$ *where* $v = (1, 3, 7)^T$ *(right eigenvector) and* $w = (1, 7, 10)^T$ *(left eigenvector, normalised so* $w^T v = 1 \pmod{13}$*).*

The two-term mod-13 formula is:

$$T_3^N = 7^N P_1 + P_{23} T_3^N \pmod{13}.$$

The projectors satisfy $P_1^2 = P_1$, $P_{23}^2 = P_{23}$, $P_1 P_{23} = 0$, $P_1 + P_{23} = I_3$, $T_3 P_1 = 7P_1$, *all exact modulo 13 and verified at startup.*

This decomposition is used at every layer of the forward pass. The Schwinger-Dyson solver (`SchwingerDysonSolver`) inverts $(I - KT_3)$ by applying Fermat's inverse $(1 - 7K)^{11}$ to P_1 channels and the matrix T_3^7 to P_{23} channels, exploiting the split exactly. The `SpectralSelfRefiner` module verifies $P_1 + P_{23} = I_3$ and $P_1 P_{23} = 0$ at every layer call, raising an error if the projectors drift.

2.2. Conformal dimension and connection to physics

The conformal dimension $\Delta = \ln \eta / 2 \approx 0.30469$ arises from the modulus of the subdominant eigenvalues: $|\lambda_2| = \eta^{-1/2}$ implies $-\log |\lambda_2| = \ln \eta / 2 = \Delta$. This quantity appears in several physical predictions derived in [1]:

$$\begin{aligned} \sin \theta_C &= |\lambda_2| \cdot \Delta = \eta^{-1/2} \cdot \Delta \approx 0.22466 && \text{(Cabibbo angle, observed: 0.22537),} \\ \sin^2 \theta_W &= \frac{\Delta}{1+\Delta}\left(1 - \frac{\Delta^2}{6}\right) \approx 0.22992 && \text{(Weinberg angle, observed: 0.23122),} \\ R_{\mathrm{QNM}} &= \arg(\lambda_2)/\Delta \approx 7.142 && \text{(QNM ratio, to be compared with LIGO data).} \end{aligned} \tag{2}$$

None of these numbers are put in by hand; they follow from $P(\eta) = 0$ alone. We include them here to establish that the constants of Zeta's architecture are the same constants that appear in parameter-free physical predictions.

3. The Unified Kernel and Ultrametric Topology

Definition 3.1 (13-adic valuation)**.** For $n \in \mathbb{Z}$, $v_{13}(n) = \max\{k \geq 0 : 13^k \mid n\}$, with $v_{13}(0) = \infty$. The *ultrametric distance* is $d_{13}(i, j) = v_{13}(|i - j|)$.

The ultrametric inequality $d_{13}(i, k) \leq \max(d_{13}(i, j), d_{13}(j, k))$ is stronger than the ordinary triangle inequality. It partitions any sequence into a hierarchical family of ultrametric balls: positions i and j are in the same ball of radius k iff $13^k \mid (i - j)$. This is the geometry of the p-adic integers $\mathbb{Z}_p$, which was studied in detail in [1] as the natural topology of the Tribonacci foliation.

Definition 3.2 (Unified kernel)**.**

$$G(i, j) = \eta^{-v_{13}(|i-j|)} \in \mathbb{Z}_{13}[\eta], \quad \eta^{-d} = \eta^{(168-d) \bmod 168}.$$

Proposition 3.3 (Uniqueness)**.** *Among kernels on* $\mathbb{Z}$ *that are (i) translation-invariant, (ii) compatible with ring multiplication:* $K(i, j) = f(v_{13}(|i - j|))$ *for some ring-valued* f*, and (iii) monotone in the ultrametric:* $v_{13}(|i - j|)$ *large implies* $K(i, j)$ *large, the choice* $K = G$ *is the unique one consistent with* $f(d) = \eta^{-d}$*.*

3.1. Specialisations across 63 modules

The power of this construction is that every module in Zeta is a specialisation of G, not an independent design choice. The table below shows the primary specialisations.

Module	Formula	Group	Purpose
`AlgebraicAttention`	$A(i,j) = G(i,j)$	Attention	Pairwise token weights
`PositionalDecay`	$w(i) = G(i,0) = \eta^{-v_{13}(i)}$	Embedding	Positional encoding
`HilbertEta`	$\langle u, v\rangle = \sum_k u_k v_k \eta^{-k}$	Spectral	Inner product
`BornNorm`	$N(c) = \sum_k c_k^2 \eta^{-k}$	Spectral	Normalisation
`FeynmanPathIntegral`	$G_0(i,j) = \eta^{-v_{13}(\lvert i-j\rvert)}$	Quantum	Free propagator
`M3to1CoarseGraining`	bond dim $\chi = 3^3 = 27$	MERA	Scale invariance
`PAdicUltrametricTree`	tree keyed by v_{13}	Cache	KV retrieval
`DensityMatrix`	$\rho_{ij} = \psi_i \psi_j$	Quantum	State entropy
`FractionalDissipation`	$\mathrm{Lap}[i,j] = G(i,j)$	Physics	UV damping
`WittVector`	depth-1 Witt poly product	Algebra	Entropy expansion

The MERA bond dimension $\chi = 27 = 3^3$ is not a hyperparameter. The value 3 is the branching factor of the Tribonacci foliation (three terms in the recurrence, three spatial dimensions), and the exponent 3 is the Witt depth, both determined by the ring structure. This matches the MERA bond dimension derived in [1] from the holographic area law of the Tribonacci tensor network.

4. Module Architecture

Zeta (formerly GeoLLM, renamed in the current release) is implemented as 63 classes organised into 13 groups. Every class uses only `torch.long` tensors. No `float32` or `float64` tensor appears anywhere in the forward pass. The `HyperparameterAbolisher` module verifies this at startup by scanning all parameters and buffers. Below we describe each group and the algebraic role of each module.

4.1. Ring and kernel primitives

The seven foundational classes implement all arithmetic directly in $\mathbb{Z}_{13}[\eta]$.

`Z13` implements all ring operations: scalar `add`, `sub`, `mul`, `inv` ($= u^{167}$ by binary exponentiation with $n = 167$), `pow_elem`; and batched versions `t_add`, `t_sub`, `t_mul`, `t_inv` for `(...,3)` Long tensors. All 167 non-zero element inverses are verified at startup.

`UnifiedKernel` wraps $G(i,j) = P_1^{-1} \eta^{-v_{13}(|i-j|)}$ with methods `matrix(L)` (builds $(L, L, 3)$ Long tensor) and `apply(x)` (ring matrix-vector product).

`T3Matrix` precomputes the 168-entry power table at startup by 167 sequential ring `(3,3)@(3,3)` matmuls. `power(n)` returns `_table[n%168]` in O(1). `spectral_power(n)` returns $7^n P_1 + P_{23} T_3^n$ using Theorem 2.5.

`PAdicNorm` implements v_{13}, the ultrametric distance d_{13}, and its verification `ultrametric_check(i,j,k)`.

`HilbertEta` provides the eta-weighted inner product $\langle u, v\rangle = \sum_{k=0}^{D-1} u_k v_k \eta^{(168-k) \bmod 168}$ over the Hilbert space H_η of ring sequences. The weight η^{-k} gives infrared dominance: low-index channels carry higher weight.

`BornNorm` computes $N(c) = \sum_k c_k^2 \eta^{-k}$ and its Fermat inverse $N^{11} \pmod{13}$, used by `AlgebraicAttention` to normalise attention scores.

`PositionalDecay` builds the $(L, L, D, 3)$ positional weight tensor from $G(i, 0) = \eta^{-v_{13}(i)}$, with a Tribonacci light-cone mask that zeroes positions outside $t^*(L)$ steps. No parameter is learned; the encoding is a deterministic function of position.

4.2. Embedding, attention, and MERA

`ZRingTokenEmbedding` maps token i and channel j to $\eta^{(7i+3j) \bmod 168}$. The factor $7 = \lambda_1$ (dominant eigenvalue) and 3 (Tribonacci branching) are the only two integers that appear. No `nn.Embedding` and no `nn.Parameter` exist in this class.

`AlgebraicAttention` implements multi-head ring attention with H heads, embedding dimension D, and sequence length L. Queries, keys, and values are computed as ring matrix products of the input with weight matrices initialised from P_1 and P_{23}. Attention scores: $A_{b,h,i,j} = \sum_d Q_{b,h,i,d} \cdot K_{b,h,j,d} \pmod{13}$. Output: $O = N^{-1} \sum_j A_{i,j} V_j$ where $N = \mathrm{BornNorm}(A)$. Only positions within the Tribonacci window contribute. No softmax, no exponential, no floating-point number.

`M3to1CoarseGraining` implements one level of MERA with bond dimension $\chi = 27$. `disentangle_pair`(s_1, s_2): applies η^{7i} phase to each pair. `coarse_grain_3to1`: combines three tokens into one using weights η^0, η^1, η^2. `mera_layer`(x): pads to a multiple of 3 and maps over all triples, reducing length L to $L/3$.

`PAdicUltrametricTree` is a KV cache with retrieval keyed by $v_{13}(|\text{query} - \text{key}|)$: the stored entry closest in the 13-adic metric. This cache structure was proposed in [1] as the natural memory hierarchy for p-adic information processing.

`SortedStateBuffer` is a priority queue sorted by `HilbertEta.norm_sq`, used by `BuchbergerTraining` to process corrections in order of algebraic significance.

4.3. Spectral theory and p-adic fields

`SpectralDecomposition` stores P_1 and P_{23} as class-level constants and exposes `verify_resolution_identity` and `verify_eigenvalue_property`.

`SpectralSelfRefiner` maintains running estimates of P_1 and P_{23} and checks orthogonality and completeness at every layer. If the checks fail (which would indicate a bug in the ring arithmetic), `refine_step` applies one η^{-1} correction.

`SchwingerDysonSolver` inverts $(I - KT_3)$: P_1 channels via Fermat scalar inversion $(1 - 7K)^{11}$; P_{23} channels via T_3^7 (the orbit on the subdominant subspace).

`FunctionalCalculus` computes $f(T_3)$ via spectral formula: $f(T_3) = f(7)P_1 + P_{23} f(T_3)|_{P_{23}}$. Supports Taylor series (`exp_taylor`) and resolvent.

`F169Element` implements arithmetic in $\mathbb{F}_{169} = \mathbb{F}_{13}[\sqrt{2}]$ with Frobenius $\sigma(a, b) = (a, -b)$, correct since $\sqrt{2}^{13} = -\sqrt{2}$ in $\mathbb{F}_{169}$.

`WittVector` implements truncated Witt vectors of depth 3 over $\mathbb{F}_{169}$, with the depth-1 product rule $W_0 = w_0 v_0$, $W_1 = w_0^{(p)} v_1 + w_1 v_0^{(p)}$, where the ghost term $p\, w_1 v_1$ vanishes as $p = 13 = 0$ in $\mathbb{F}_{169}$. When entropy saturates ($S \geq S_G^{\max} = 10$), `open_next` opens a new Witt component, implementing a p-adic spiral.

4.4. Lie and conformal algebras

`VirasoroAlgebra` with $c = 10$ implements $[L_m, L_n] = (m-n)L_{m+n} + \frac{c}{12} m(m^2 - 1)\delta_{m+n,0}$. The central term uses $\mathrm{inv}(12) = 12 \pmod{13}$ (since $12^2 = 144 \equiv 1$); integer floor division $\lfloor \cdot/12 \rfloor$ would be algebraically incorrect. The central charge $c = 10$ comes from the mode-counting derivation in [1] of the critical string dimension from the Tribonacci foliation.

`KacMoodyAlgebra` with rank 2 implements $[J_m^a, J_n^b] = f^{abc} J_{m+n}^c + k\, m\, \delta^{ab} \delta_{m+n,0}$.

`sl2RAlgebra` implements the two-dimensional representation $L_0 = \left(\begin{smallmatrix} 7 & 0 \\ 0 & 6 \end{smallmatrix}\right)$ ($7 = \mathrm{inv}(2) \pmod{13}$), $L_+ = \left(\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right)$, $L_- = \left(\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right)$. All three commutator relations are verified mod 13.

4.5. Clifford algebra, Hopf algebra, and quantum groups

`CliffordAlgebraZ13` implements $\mathrm{Cl}(1,2)$ in the Weyl basis with Minkowski signature $(+,-,-)$ over $\mathbb{Z}_{13}$: $\gamma_0^2 = +I$, $\gamma_1^2 = \gamma_2^2 = -I$, $\{\gamma_\mu, \gamma_\nu\} = 0$ for $\mu \neq \nu$ (all verified). The matrices are $\gamma_0 = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 12 \end{smallmatrix}\right)$, $\gamma_1 = \left(\begin{smallmatrix} 0 & 5 \\ 5 & 0 \end{smallmatrix}\right)$, $\gamma_2 = \left(\begin{smallmatrix} 0 & 1 \\ 12 & 0 \end{smallmatrix}\right)$. The choice $5^2 = 25 \equiv 12 \equiv -1 \pmod{13}$ and $1 \cdot 12 = 12 \equiv -1$ ensures the correct signature.

`HopfAlgebra` on $\mathbb{Z}_{13}[S_3]$ with group-like coproduct $\Delta(g) = g \otimes g$, antipode $S(g) = g^{-1}$, and counit $\varepsilon(g) = 1$. The `comultiply` method expects a $(6,)$ Long vector of S_3 coefficients; in the forward pass this is constructed from the ring element by placing a_0, a_1, a_2 in positions $0, 1, 2$ and zero-padding to length 6.

`QuantumGroup` implements $U_q(\mathfrak{sl}_2)$ at $q = \eta$ ($\eta^{168} = 1$, so q is a 168th root of unity). The R-matrix gives $R(v \otimes w) = q^{h_v h_w}(w \otimes v)$ via `ETA_POWERS[`$(v_{\mathrm{sum}} \cdot w_{\mathrm{sum}}) \bmod 168$`]`.

`ExteriorAlgebra`: $\mathrm{wedge}(v, w) = \sum_{i<j} (v_i w_j - v_j w_i) \pmod{13}$. Anti-symmetry $\mathrm{wedge}(v, v) = 0$ is verified by the symmetric polynomial identity.

`TensorAlgebra`: tensor, symmetric ($\times 7 = \times \mathrm{inv}(2)$), and exterior products on ring element lists.

`JordanAlgebra`: $A \circ B = 7(AB + BA) \pmod{13}$. In the forward pass: $T_3^k \circ T_3^k$, with row 0 of the result added as a residual to channel 0 of the representative token.

4.6. Operator algebras and non-commutative geometry

`CStarAlgebra`: involution $A^* = \sigma(A^T)$ where $\sigma(a_0, a_1, a_2) = (a_0, -a_1, -a_2) = (a_0, 13 - a_1, 13 - a_2) \pmod{13}$ is the Frobenius automorphism of $\mathbb{Z}_{13}[\eta]$. This is the ring analogue of complex conjugation.

`VonNeumannAlgebra`: commutant $[A, B] = AB - BA \pmod{13}$, trace, modular operator $\Delta(a) = \eta^{-1} a$, and the Connes cocycle $\sigma_t(a) = \eta^t a \eta^{-t}$.

`NCGeometry`: spectral triple $(\mathcal{A}, H_\eta, D)$ where $D = \mathrm{diag}(1, 2, 3)$ (the ring Dirac operator, dimension 3×3). The commutator $[D, a] = Da - aD$ applied with $a = T_3^N$

(matching the dimension of D) gives the Connes distance formula in the Tribonacci geometry.

4.7. Quantum physics modules

`VertexAlgebra`: $Y(a,z)_n = z^{-n-1}a \pmod{13}$, where $z^{-1} = z^{11} \pmod{13}$ by Fermat. In the forward pass: $z = \text{layer} + 1$ (nonzero by construction), $n = \text{layer_idx}$.

`FeynmanPathIntegral`: free propagator $G_0(i,j) = \eta^{-v_{13}(|i-j|)}$, one-loop correction $G^{(1)} = G_0 \cdot T_3^N \cdot G_0 \pmod{13}$. This matches the holographic path integral formulation in [1].

`TribitQuantumSystem`: 3-level qudit (basis $|0\rangle, |1\rangle, |2\rangle$) with S_3 Galois symmetry. `decompose_galois` decomposes any ring state into S_3 irreducible representations.

`DensityMatrix`: $\rho_{ij} = \psi_i \psi_j$, von Neumann entropy proxy $S = \text{Tr}(\rho) - \text{Tr}(\rho^2) \pmod{13}$, and bipartite entropy by partial trace.

`GeometricHamiltonian`: $H = T_3 - I$, energy spectrum $E_k = (7^k - 1) \pmod{13}$ for $k = 0, \ldots, 12$. Ground state: $E_0 = 6$ on the P_1 eigenspace.

`BerryPressure`: geometric phase classification by orbit proximity. `forward`(x) returns $\text{argmin}_c \left|(x)_{a_0} - c\right|$ over class centres $c \in \{0, \ldots, 3\}$, as a Long scalar.

4.8. Perturbation theory and thermodynamics

`TimeIndependentPerturbation`: first-order correction $E^{(1)} = \langle n|V|n\rangle / \langle n|n\rangle$ using the H_η inner product.

`DysonSeries`: time evolution $U = \sum_{k=0}^{3} T_3^k \eta^k / k! \pmod{13}$, where $k! \pmod{13}$ is computed by Fermat. Returns a $(3,3)$ Long evolution matrix.

`VariationalPrinciple`: variational bound $E_0 \leq \langle\psi|H|\psi\rangle / \langle\psi|\psi\rangle$, saturated by $v = (1,3,7)$ with $E_0 = 6$.

`ETHModule`: eigenstate thermalisation; modular Hamiltonian $K = (I - \rho)\rho \pmod{13}$.

`GeometricCrossEntropy`: $L_k = (s_{\text{pred}} - s_{\text{bio}} + 7\, s_{\text{bio}}) \pmod{13}$.

`FreeEnergyFunctional`: $F_k = (11 + L_k) \pmod{13}$, where $11 = \text{Tr}(T_3) - \text{Tr}(I) = 1 - 3 = -2 \equiv 11$. $F_k = 0$ is the equilibrium condition.

`ThermodynamicOptimizer`: optimal temperature $T_{\text{opt}} = S_G^{\max} \cdot \lambda_1 = 10 \times 7 = 70 \equiv 5 \pmod{13}$.

`EntropySaturation`: hard cap $S_G^{\max} = 10$, derived from the coefficient sum of $\eta^3 = \eta^2 + \eta + 1$: $1 + 1 + 1 = 3$, cap $= 13 - 3 = 10$. When $S \geq 10$, opens a new Witt level.

`FractionalDissipationOperator`: p-adic Laplacian $\text{Lap}[i,j] = G(i,j)$ for $i \neq j$, $\text{Lap}[i,i] = -\sum_{j \neq i} G(i,j) \pmod{13}$. Each row sums to zero (verified). Applied at every layer to damp UV modes on the Tribonacci lattice.

4.9. Training algorithms

`BuchbergerTraining`: the central training module, described in detail in Section 5. Also implements `buchberger_algorithm` with full F5 criterion, `is_unit_ideal`,

`bezout_identity`, and `quantum_rollback`.

`HenselLifter`: Newton-Hensel lifting of $\lambda_1 = 7$ to precision 13^k, verified for $k = 1, 2, 3$.

`S3GaloisSymmetry`: the Galois group $S_3 = \mathrm{Gal}(\mathbb{Q}(\eta)/\mathbb{Q})$, which permutes the three eigenvalues $\{\lambda_1, \lambda_2, \lambda_3\}$. Applied with the identity permutation in the forward pass to register the symmetry path.

`PageCurve`: tracks $S(N)$ across layers, verifies the shape S rises then falls (Page curve), and records compression rate $(S_{\max} - S_{\text{final}})/S_{\max}$. The Page time $t_{\text{Page}}/N = 1/2$ is exact by time-reversal symmetry [1].

`GeometricUncertainty`: Heisenberg bound $\delta_A \cdot \delta_B \geq |[A, B]| \cdot \mathrm{inv}(2) = |[A, B]| \cdot 7$.

`LiouvilleVolumeGuard`: asserts $\det(T_3^k) = 1$ at every layer. This is Corollary **??**; a failed assertion would indicate a ring arithmetic bug.

4.10. DNA topology modules

`TopoisomeraseSurgery`: $L_- = \left(\begin{smallmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{smallmatrix}\right)$, $L_-^3 = 0$ (nilpotent index 3, verified). Surgery: $u \mapsto u + L_-^2 u \pmod{13}$. The linking number change $\delta_{\text{Lk}} = -2 \equiv 11 \pmod{13}$ models Topoisomerase II, which changes Lk by -2 per catalytic cycle. The same ring arithmetic that governs token prediction governs DNA topology.

`TopologicalMalignity`: detects when $\mathrm{GB}(x - s_{\text{pred}}, x - s_{\text{bio}}) = \{1\}$ (contradiction ideal, equivalent to wrong prediction). Also classifies ring-valued Jones polynomials against a healthy library.

`TopologicalLock`: enforces the structural invariant $\alpha = \eta$, raising `TopologicalLockError` if any modification is attempted.

`DNATokenizer`: encodes nucleotide sequences as codons (3-mers) indexed $0, \ldots, 63$, mapped to η^{codon} via the embedding.

4.11. System integrity and tokenisation

`SystemIntegrityGuard`: noise injection and isolation check.

`AdelicVerifier`: verifies all ring axioms at startup ($\eta^3 = \eta^2 + \eta + 1$, $\det(T_3) = 1$, $T_3^{168} = I_3$, all 168 inverses, $P_1^2 = P_1$, Clifford relations, $\mathfrak{sl}_2(\mathbb{R})$ brackets).

`HyperparameterAbolisher`: scans all parameters and buffers for `float32` or `float64` tensors; raises `HyperparameterError` if any are found. Also checks for gradient-based optimisers.

`InferenceEngine`: sanitises (clamps to $[0, 12]$), maintains the KV cache, performs MERA chunking on long sequences, and inserts into both the ultrametric tree and the sorted buffer at every step.

`GeneralTokenizer`: maps text to $\mathrm{ord}(c) \bmod 168$, DNA sequences to codon indices, and physics particles to a fixed symbol table.

`ProductionScalingTest`: benchmarks $t^*(L)$ for $L \in \{64, 128, 256, 1024\}$ and verifies the $O(\ln L / \ln \eta)$ scaling claim.

5. Training by Polynomial Algebra

Theorem 5.1 (Buchberger-Nullstellensatz training)**.** *Let* $s_{\text{pred}}, s_{\text{bio}} \in \mathbb{Z}_{13}$*. The ideal* $I = (x - s_{\text{pred}}, x - s_{\text{bio}}) \subset \mathbb{Z}_{13}[x]$ *satisfies:*

(i) If $s_{\text{pred}} = s_{\text{bio}}$*:* $I = (x - s_{\text{pred}})$*, no correction.*

(ii) If $s_{\text{pred}} \neq s_{\text{bio}}$*:* $I = (1)$ *(unit ideal).* $\text{GB}(I) = \{1\}$*. Bezout coefficients:* $\delta = (s_{\text{bio}} - s_{\text{pred}})^{-1} = (s_{\text{bio}} - s_{\text{pred}})^{11} \pmod{13}$*. Weight update:* $W \mathrel{+}= \delta \cdot \eta^k \pmod{13}$.

The correction magnitude δ forms a Latin square over $\mathbb{Z}_{13}$: every nonzero error $(s_{\text{bio}} - s_{\text{pred}}) \in \{1, \ldots, 12\}$ has a unique inverse. The map $(s_{\text{pred}}, s_{\text{bio}}) \mapsto \delta$ is a bijection on $\mathbb{Z}_{13} \setminus \{0\}$, guaranteeing that no two different errors produce the same correction.

5.1. Free energy and equilibrium

The ring cross-entropy $L_k = (s_{\text{pred}} - s_{\text{bio}} + 7s_{\text{bio}}) \pmod{13}$ and free energy $F_k = (11 + L_k) \pmod{13}$ define a thermodynamic framework. The constant $11 = -2 \equiv \text{Tr}(T_3) - \text{Tr}(I_3) = 1 - 3 \pmod{13}$. The optimal temperature is $T_{\text{opt}} = S_G^{\max} \cdot \lambda_1 = 10 \times 7 \equiv 5 \pmod{13}$. Equilibrium $F_k = 0$ is equivalent to exact prediction. This framework connects to the holographic thermodynamics of [1], where the Hawking temperature is derived from the same polynomial $P(\eta) = 0$.

6. Theoretical Properties

6.1. Scrambling time

Let $T(k)$ denote the Tribonacci sequence: $T(0) = T(1) = 1$, $T(2) = 2$, $T(k) = T(k-1) + T(k-2) + T(k-3)$. The growth rate is $T(k) \sim C\eta^k$ with $C > 0$.

Theorem 6.1 (Scrambling bound)**.** $t^*(L) = \min\{k \in \mathbb{N} : T(k) \geq L\} = O(\ln L / \ln \eta)$*. Explicit values:* $t^*(128) = 9$ *(*$T(9) = 149$*),* $t^*(256) = 10$ *(*$T(10) = 274$*),* $t^*(1024) = 13$ *(*$T(13) = 1705$*). After* $t^*(L)$ *layers, every pair of positions is connected by* G*.*

This matches the Hayden-Preskill fast-scrambling bound [6], derived in a black hole context: a system of entropy S scrambles in $O(S/\ln S)$ steps. In Zeta, L plays the role of entropy and the logarithm base is η. The bound was derived in [1] from the OTOC (Out-of-Time-Order Correlator) and Lyapunov exponent of the transfer matrix.

6.2. Spectral gap and information growth

The complete three-term spectral formula (Theorem 2.5) gives

$$T_3^N = \eta^N P_1 + \eta_2^N P_2 + \eta_3^N P_3,$$

with $|\eta_2| = |\eta_3| = \eta^{-1/2} \approx 0.73735$. The ratio of the dominant to subdominant norm grows as $\eta^N/|\eta_2|^N = (\eta \cdot \eta^{1/2})^N = \eta^{3N/2}$. At $N = 10$: $\eta^{15} \approx 7.28 \times 10^3$. At $N = 20$: $\eta^{30} \approx 5.30 \times 10^7$. After $O(1)$ layers, the P_{23} component of any state is algebraically negligible and the dynamics are governed entirely by the rank-one projector P_1.

In the mod-13 implementation this is made exact: the Schwinger-Dyson solver inverts the P_1 channel by a single Fermat scalar $(1-7K)^{11} \pmod{13}$, with no iteration or approximation, while the P_{23} channel receives the deterministic ring rotation T_3^7. The two channels are handled by completely different algebraic operations, reflecting their fundamentally different spectral characters.

6.3. Entropy bound and Witt expansion

Proposition 6.2 (Entropy cap)**.** *The entropy proxy* $S(\rho) = \mathrm{Tr}(\rho) - \mathrm{Tr}(\rho^2) \pmod{13}$ *satisfies* $S \leq S_G^{\max} = 10$.

When $S \geq 10$, the `WittVector.open_next` method opens a new Witt component, implementing a p-adic spiral. The depth-1 Witt product $W_1 = w_0^{(p)} v_1 + w_1 v_0^{(p)}$ involves the Frobenius ${}^{(p)} : (a, b) \mapsto (a, -b)$ of $\mathbb{F}_{169}$, which is the ring-theoretic analogue of complex conjugation at the prime 13.

6.4. Lyapunov exponent and scrambling

The connection to chaos is through the OTOC in the Tribonacci foliation. The Lyapunov exponent of the transfer matrix is $\lambda_L = \ln \eta \approx 0.60998$, which is the same quantity that appears in the conformal dimension $\Delta = \lambda_L/2$. This is not a coincidence: the Maldacena-Shenker-Kitaev bound [1] gives the maximum Lyapunov exponent as $2\pi T$, and the identification of T with the Hawking temperature of the holographic dual gives $\lambda_L = \ln \eta$ without any free parameter.

6.5. Volume preservation and unitarity

Corollary **??** states $\det(T_3^N) = 1$ for all N. This is the ring analogue of Liouville's theorem in Hamiltonian mechanics: the phase space volume is preserved under time evolution. In the holographic interpretation [1], it corresponds to the unitarity condition $\det(T_3) = 1$ derived from the requirement that the Tribonacci foliation be a unitary evolution.

7. Physical Connections

The modules of Zeta are not independent inventions; they are the ring implementations of structures derived in [1] from the single polynomial $P(\eta) = 0$. The prior work established, among other results, the following quantities from $P(\eta) = 0$ alone or from the Tribonacci hypothesis:

Quantity	Tribonacci formula	Predicted	Observed	Zeta module
Conformal dim. Δ	$\ln \eta/2$	0.30469	— (exact)	`FunctionalCalculus`
Cabibbo angle $\sin\theta_C$	$\eta^{-1/2} \cdot \Delta$	0.22466	0.22537	`F169Element`
Weinberg angle $\sin^2\theta_W$	$\frac{\Delta}{1+\Delta}(1 - \frac{\Delta^2}{6})$	0.22992	0.23122	`VirasoroAlgebra`
QNM ratio R	$\arg(\lambda_2)/\Delta$	7.142	(LIGO, TBD)	`GeometricHamiltonian`
Page time	$t_{\text{Page}}/N = 1/2$	exact	—	`PageCurve`
Scrambling	$t^* = \log S/\log \eta$	verified	—	`ProductionScalingTest`
MERA bond dim	$\chi = 3^3 = 27$	exact	—	`M3to1CoarseGraining`
Entropy cap	$S_G^{\max} = 10$	exact	—	`EntropySaturation`
Virasoro c	$c = 10$ (mode count)	exact	—	`VirasoroAlgebra`
Topol. surgery δ_{Lk}	$-2 \equiv 11 \pmod{13}$	exact	TopII	`TopoisomeraseSurgery`

All numbers in the "Predicted" column are drawn from [1] and verified by SymPy in that document; they are reproduced here for context. The Zeta architecture was designed so that the algebraic identities underlying each prediction are implemented as executable code.

8. Validation Status and Code Availability

8.1. Algebraic verification

The 30 axioms verified at startup constitute the current validation baseline. These include all projector identities, the Clifford anticommutators, the Hensel lifting chain, the Tribonacci scrambling values, and the Virasoro central coefficient. They are verified in pure Python arithmetic with no external dependencies. A failed assertion before inference indicates a ring arithmetic bug; none have been observed in the current implementation.

8.2. Planned empirical validation

Zeta has not yet been trained on any real dataset. The following validation experiments are in preparation.

(1) **Scrambling benchmark.** Measure the empirical scrambling time as a function of L and compare to the theoretical bound $t^*(L) = O(\ln L/\ln \eta)$. This test requires no training data and can be run on random ring-valued sequences.

(2) **DNA sequence prediction.** Apply `DNATokenizer` to standard genomic benchmark datasets and measure prediction accuracy using the Buchberger training rule. We expect this to be the first nontrivial empirical test, since the DNA codon structure (4 nucleotides, codons of length 3, 64 total) maps naturally into the ring arithmetic.

(3) **Comparison with quantised transformers.** Compare Zeta against standard transformer models quantised to 4-bit integer arithmetic on the same tasks. This comparison isolates the effect of the algebraic architecture from the effect of reduced numerical precision.

(4) **Physical prediction verification.** The QNM ratio $R_{\mathrm{QNM}} = \arg(\lambda_2)/\Delta \approx 7.142$ predicted in [1] is to be compared against LIGO ringdown data. This is not a test of Zeta as a language model, but a test of the underlying algebraic framework.

We note that the validation tests described above differ qualitatively from the validation of conventional neural networks. For Zeta, the primary validation is algebraic: all 30 axioms either hold or do not hold, with no approximation. Empirical validation on natural language tasks is secondary and subject to the open question of whether the ring arithmetic confers any practical predictive advantage over floating-point baselines.

8.3. Code availability

The full source code of Zeta, comprising 4852 lines and 63 classes, is available at Zenodo record 18937195 [1]. A cleaned release on GitHub, with documentation, unit tests, and a minimal training loop using the Buchberger rule on synthetic data, is currently in preparation and will be linked from the Zenodo page upon completion. The code requires `Python` $\geq$ 3.10 and `PyTorch` $\geq$ 2.0. No GPU is required; all operations are `torch.long` on CPU.

9. Limitations

9.1. Implementation limitations

No empirical training. The Buchberger correction rule has been implemented and algebraically verified. No training run on a real dataset has been executed. Whether the rule converges to a useful predictor on natural language is an open empirical question.

Output representation. The output logit is a ring element $(a_0, a_1, a_2) \in \mathbb{Z}_{13}^3$, not a probability vector. Mapping to standard evaluation metrics requires a decoding convention, which has not yet been standardised.

CPU throughput. Long integer matrix multiplication on CPU is not accelerated by the same hardware as floating-point GEMM. A direct throughput comparison with floating-point transformers would not be fair without custom ring arithmetic hardware.

Tokenisation. The current tokeniser maps characters to $\mathrm{ord}(c) \bmod 168$. Subword tokenisation, which would require a ring-valued BPE algorithm, is not implemented.

9.2. Open mathematical problems

Convergence of Buchberger training. The training rule is exact per step but convergence to a global minimum of F_k has not been established.

Generalisation to other primes and polynomials. The factorisation criterion that singles out $p = 13$ for the Tribonacci polynomial can in principle be applied to other sequences and other primes. The resulting hierarchy of rings has not been classified.

Ring-valued probability theory. The Born norm $N(c) = \sum_k c_k^2 \eta^{-k}$ defines a ring-valued quadratic form. A full measure-theoretic framework over $\mathbb{Z}_{13}[\eta]$ has not been developed.

Acknowledgements

The author thanks the Zenodo platform for open preprint hosting and acknowledges that all numerical results in this paper and in [1] were verified by SymPy without exception.

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